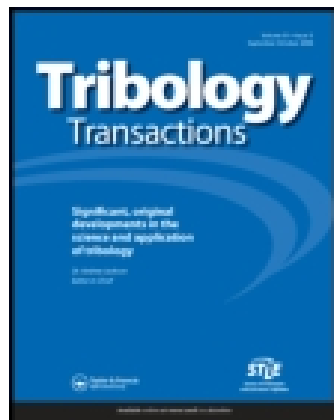




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On Competing Failure Modes in Rolling Contact

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On Competing Failure Modes in Rolling Contact

By T. E. TALLIAN¹

Rolling contact failures are classified according to their failure mode as (a) wear, (b) plastic flow, (c) fatigue, and (d) bulk failures, with the last class arising outside the immediate contact area. The wear failures are subdivided into mild wear by loose particle removal and smearing involving metal transfer. Plastic failures may arise due to overload or to temperature imbalance. Fatigue is of the spalling or the surface distress type. An analysis of the stress conditions in a rolling contact is sketched; the stepwise refinements of the contact model are considered, starting from Hertz theory, and progressively including surface traction, plasticity, elastohydrodynamic lubricant films, surface microtopography, and the inhomogeneities of real metals. Each failure mode is associated with the relevant severity parameters of the contact stress condition, and a description of its mode of formation is given. Guidelines are provided for the identification of that failure mode which, among competing modes, is most likely to render a rolling contact inoperative in a given operating environment.

Nomenclature

- w = Wear rate
- ξ_0 = Elastohydrodynamic film thickness-roughness ratio
- a = Area of real asperity contact
- a_0 = Area of Hertz contact
- K = Rolling wear constant
- h_0 = "Uniform" elastohydrodynamic film thickness
- $\sigma_{1,2}$ = rms roughness of the contacting surfaces
- W = Load
- S = Sliding velocity
- ϵ = Constant exponent (smearing load vs. speed equation)
- P = Probability to failure
- τ_0 = Maximum alternating shear stress range
- Z_0 = Depth below surface, of stress τ_0
- N = Number of stress cycles
- e = Weibull dispersion parameter
- V = Volume (of material under stress)
- f = Function sign
- h, c = Exponents in life equation

Introduction

CONSIDERABLE advances have been recently made in the analysis and understanding of rolling contact failure mechanisms. It is now possible to outline a reasonably coherent view of these failure modes, which supersedes, in several respects, the customary contact failure classification as contained, for example, in (1). It is well known that there are several competing failure modes in a rolling contact. It appears possible to isolate the operating conditions which favor a given failure mode over others, and thereby

predict the useful life of the contact in a more realistic way than was possible before. One can then take engineering action to control premature or avoidable rolling contact failures. One can also more readily recognize "state of the art" limitations, where they exist, and thereby reduce futile experimentation.

This paper represents one author's view of the current, highly quantitative state of a unified rolling contact failure model, and encompasses the several competing failure modes so far recognized. It is subdivided into the following sections:

- I. Description of competing rolling contact failure modes.
- II. Survey of engineering parameters influential in causing contact failure by any failure mode.
- III. Analysis of rolling contact failures, aimed at relating failure mechanism to the engineering parameters of the contact.

Some of the failure descriptions in Section I are classical; others have been redefined. Failure parameters are listed only very briefly in Section II (a much more detailed coverage of this subject will be published elsewhere²). Most, if not all, of these parameters are of recognized influence, as shown by reference to the work cited in footnote 2. The interpretations in Section III are in part taken from the literature; in these cases they are referenced. Many of the interpretive descriptions, however, are presented for the first time. As any failure model, they are probably incapable of absolute proof. They are offered because they appear useful in coordinating current knowledge, and, hopefully, can be refined to accommodate the better knowledge of the future.

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² Tallian, T. E., "Rolling Contact Failure Control through Lubrication," Conference on Lubrication and Wear, Paper 14. Institute of Mechanical Engineers, London (1967).

I. Failure modes of rolling contacts

A rolling contact made of hard steel fails in almost all practical cases by one of the mechanisms shown in Table 1.³ Of the four main groups of failure modes

TABLE 1
Failure Modes of Rolling Contacts

Mode	Manifestation
1. Wear type failures	1.1 Surface removal 1.1.1 Removal of loose particles ("wear") 1.1.2 Chemical or electrical surface removal 1.2 Cumulative material transfer between surfaces ("smearing")
2. Plastic flow	2.1 Loss of contact geometry due to cold flow 2.2 Destruction by material softening due to unstable overheating
3. Contact fatigue	3.1 Spalling 3.2 Surface distress
4. Bulk failures	4.1 Overload cracking 4.2 Overheat cracking 4.3 Bulk fatigue 4.4 Fretting of fit surfaces 4.5 Permanent dimensional changes

listed, the first three are confined, at least initially, to the material volume subject to contact stress. The fourth group collects the "noncontact type" failure modes such as bulk cracking, fretting of fit surfaces, permanent dimensional changes of the material, overall rust, etc. Comprehensiveness of this last group is not attempted here.

FAILURE MODE 1. WEAR TYPE FAILURE

All solid surfaces in moving contact, including rolling contacts, undergo wear unless totally separated by lubricant. Generally, the true rolling contacts in a bearing wear so little that detection of removed material requires sensitive techniques such as radio-tracer analysis. *This* wear does not lead to failure. Nonetheless, the following two types of wear failure occur in rolling contact.

A. Wear failure by removal of loose wear particles ("wear"). Significant amounts of material can be removed from Hertzian contact surfaces if abrasive contaminants are present in the contact area, if there is highly inadequate lubrication under mild operating conditions, or if the contact endures high rates of sliding, again under relatively mild operating conditions. This failure mode is characterized by the appearance of voluminous amounts of wear particles in the lubricant and abrasion of the contact surface. Generally, the surface character is impaired only

³ A comprehensive listing of the visual appearance of contact failures is given by Kaufman and Walp in (1).

gradually, and in some rolling bearings, considerable wear can occur before they become unfit for operation. Diagnosis of wear failure can be accomplished by geometrical or weight measurement of the material removed or by observation of loose wear particles near the contact.

Failures similar in effect but different in origin arise from corrosive lubricants and from the passage of electric current across Hertz contacts. In both cases, surface material is removed, but it is not recoverable by filtration from the lubricant. Diagnosis is possible from surface appearance (1).

Figure 1 illustrates the geometrical consequences of wear on a ball. A surface trace across the rolling track shows two ruts formed near the edges. These ruts were formed by material removal. (The original surface gives a straight profile trace because the ball curvature has been compensated for.)

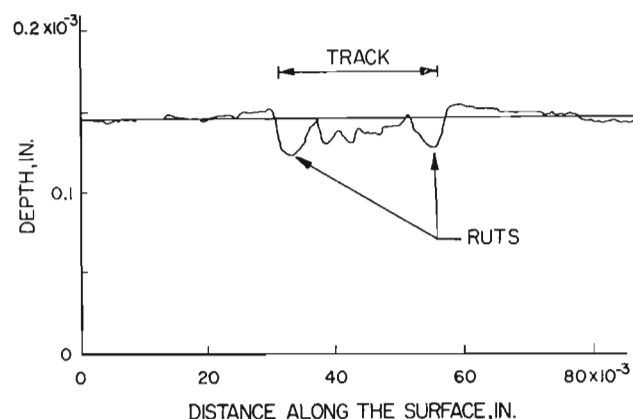


FIG. 1. Profile tracing across wear track on a ball (redrawn).

B. Cumulative material transfer between surfaces ("smearing"). A failure phenomenon described in engineering as "galling" or "smearing" takes place occasionally in *nominally* rolling Hertz contacts. It proceeds by the formation of "welded" junctions between the opposing surfaces in contact with subsequent junction growth of a cumulative nature such that, after a generally short period, macroscopic quantities of material have been transferred onto one or both of the rolling surfaces. Concurrently, depressions may form from which the transferred material was removed.

In the more advanced stages, the smearing phenomenon is detectable by visual observation of the surface. At the onset, metallographic observation, or autoradiography of transfer particles coming from a radioactive rolling element, has higher resolution.

Smearing failures appear to be confined to contacts which have undergone considerable sliding. Smearing is generally a self-aggravating condition, and if allowed to progress, often is followed by a heat imbalance failure due to the high friction losses gener-

ated between the smeared surfaces. If the operating conditions which have led to smearing are relaxed, the smearing failure may be arrested and there will be some plastic smoothing and possibly a wearing away of the transferred material which may permit continuing operation. In Fig. 2, metal transfer (smeared



FIG. 2. Metal transfer (smearing) on a sliding ball showing sheared metal on the track. Direction of sliding is from left to right.

ing) observed in a *sliding* ball is shown in a low magnification micrograph. The transferred metal appears as a bright, fracture-like patch marked in the figure.

FAILURE MODE 2. PLASTIC FLOW

Almost all rolling elements are made of hard metals, predominantly steel (a few are of nonferrous "superalloys," cermets, ceramics, or plastics). If a rolling Hertz contact made of metal is overloaded, there will be plastic flow, which, if severe enough, will destroy the intended contact geometry by forming "brinell marks" if the load is stationary, or by distorting the rolling track if the contact rolls. Plastic flow on a small scale occurs under quite reasonable contact loads (2). This has been recognized by defining a "static capacity" for rolling bearings, not by

requiring the absence of all plastic deformation but by limiting it to a tolerable magnitude. Figure 3 is a high magnification surface profile trace across a ball track which underwent plastic flow. Without flow, the trace would be a straight horizontal line because the original curvature of the surface has been compensated for. In the center of the track, a depression was formed, and material was pushed above the original surface near the track edges. This latter feature shows that the phenomenon was plastic flow and not wear (although plastic flow can exist without pushed-up ridges).

The diagnostic methods for observing plastic flow failure in a rolling contact are generally geometrical, and the distortion of contact geometry is measured.

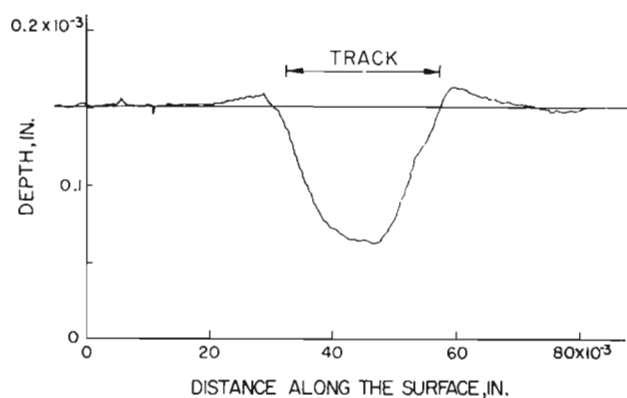


FIG. 3. Profile tracing across plastically deformed track on a ball (redrawn).

The primary effect of plastic flow on the functionality of a mechanism is the geometric distortion, which impairs accuracy, increases looseness in a bearing, or has other undesirable consequences. Other influences can arise from the effects of cold working on the material, but these are rarely catastrophic.

An essentially similar, but much more destructive, plastic flow phenomenon is observed in rolling bearings if their thermal balance becomes unstable by generation of more heat in the bearing than is being removed. The temperature of the parts increases, soon leading to the destruction of the lubricant, higher frictional losses, progressively stronger heating, eventual softening of the contact material, deformations causing even higher frictional losses, and soon, to the total destruction of the mechanism. Figure 4 illustrates this thermal imbalance type of failure in a ball bearing. The gross destruction is always easy to recognize because it appears predominantly plastic as in hot forming, and therefore dissimilar from the disintegration by fracture which can occur as a result of cracking failure. Very often, temperature imbalance failures are so grave that little is left of the mechanism from which to identify a failure source.

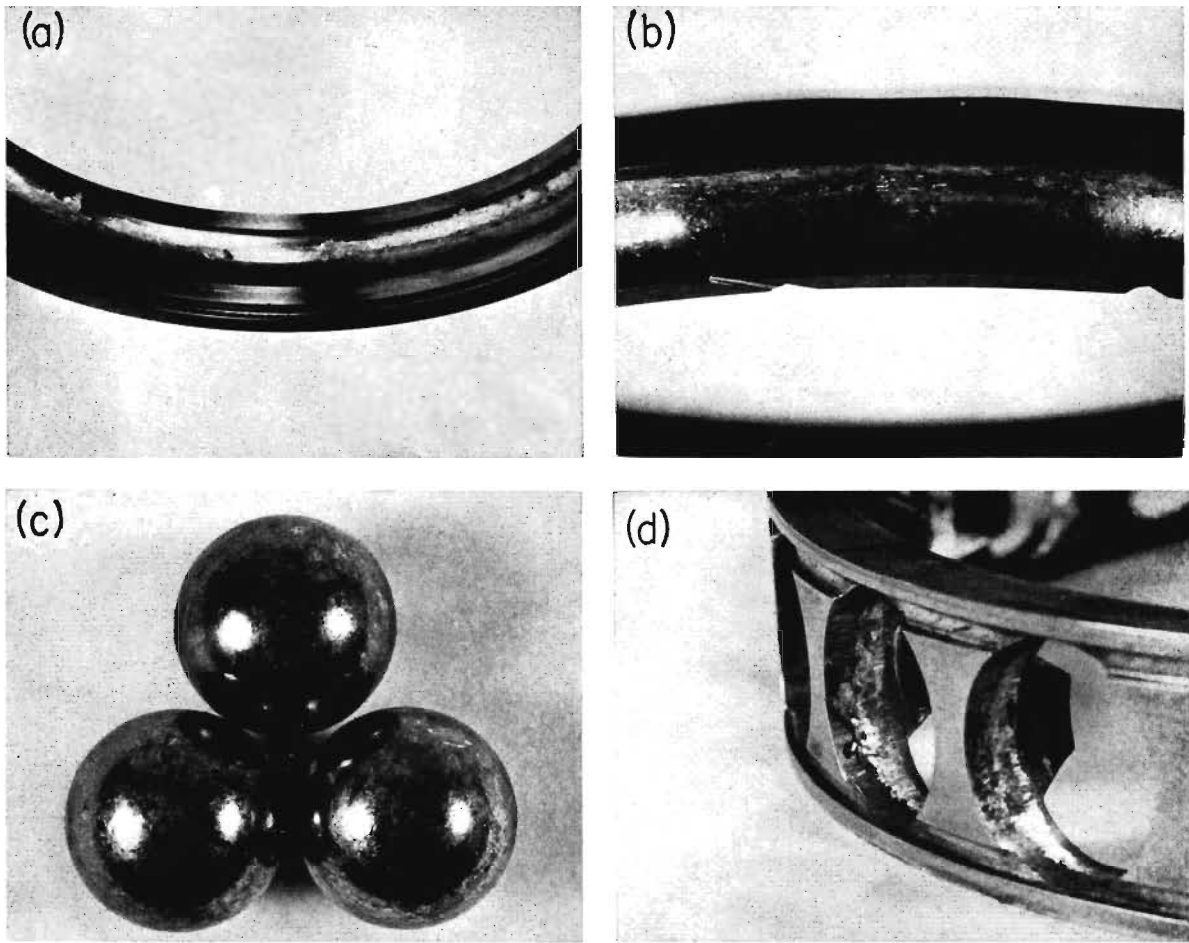


FIG. 4. Parts of rolling bearing destroyed by heat imbalance failure. (a) Outer ring with transferred, foil-like metal in groove. (b) Inner ring showing severe plastic working of groove. (c) Plastically worked balls. (d) Silver-plated separator with plating worn through in the pockets.

FAILURE MODE 3. CONTACT FATIGUE

The classical failure mode of a rolling contact between hard steel surfaces is contact fatigue. [Similar failures have been reported in ceramics (3, 4).] One type of such fatigue has been extensively described and is designated as spalling fatigue. (This is, at times, also referred to as pitting or flaking failure, but either terminology can lead to confusion with other failure modes and is avoided in this paper.)

There is also another, much less recognized, mode of failure believed to be contact fatigue, for which there is no universally accepted designation and which will be referred to as surface fatigue or (without prejudging its mechanism) as surface distress.

A. Spalling fatigue. Figure 5 is a low magnification photograph of a typical fatigue spall. It represents a crater-like depression in the original rolling surface, and its walls and bottom are formed by fracture surfaces. It is sharp edged, generally steep walled, and more or less flat bottomed, unless these features have been obliterated by further rolling-over. In the

more recent literature, subclasses of fatigue spalls have been described with reference to their presumed origin (5, 6). Subsurface originated spalls, point surface-originated spalls, and line surface-originated spalls have been mentioned. In the present paper, these distinctions will not be made, and all spalling failures will be treated together because usable criteria for distinction of any subdivisions are only now being defined; they are controversial and probably not generally applicable. If accessible for inspection, a spalling failure is immediately recognizable by the appearance of the spall. Difficulties of recognition arise only if the spall has been obliterated by further destruction of the part, or if it has not progressed beyond an extremely small magnitude, both of which problems are irrelevant to a discussion of typical failures.

B. Surface distress is a term applied to the phenomenon illustrated in Figs. 6 and 7, which show micrographs of rolling surfaces after operation. When a surface-distressed rolling surface is visually ex-

amined, the first sign observed is a "burnished" appearance (which is almost impossible to reproduce in a photograph). It is characterized by a high gloss of the metal and partial or total obliteration of the original finishing marks. It is now believed that this appearance arises from a plastic deformation of the asperities as illustrated by comparison of the replica micrographs of Fig. 6b and 6c, taken from (7), which show a surface before and after it has been rolled over a large number of times.

In a more advanced stage of this failure, small pits form on the burnished surface, which are at times aligned along ridges of the original asperities. This pitting is illustrated in Fig. 6a, where the pits appear as randomly distributed black dots on a background of finishing marks (vertical lines). The fact that these black dots really are pits, i.e., shallow depressions, is

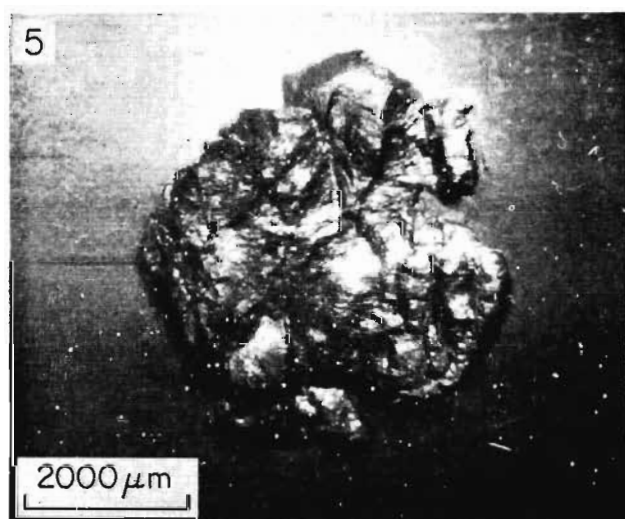


FIG. 5. Fatigue spall.

not apparent from the figure, but can be verified by focusing on their bottom with the microscope, which requires an elevation of the specimen a few micrometers toward the objective lens. Finally, an advanced stage of surface distress gives the microscopic appearance shown in Fig. 7, where transverse surface cracks have formed along three deep scratches in the original surface finish. As yet, there is no absolute proof that the black "fishbone" structures shown in Fig. 7 are indeed cracks, but such proof should now be possible by using electron fractography. It is currently surmised that they are cracks because no other reasonable explanation for their appearance has been found.

As a result of the pitting and cracking shown in Fig. 7, and because there is no appreciable material removal involved, the failure phenomenon described here is believed to be a type of surface fatigue. Its formation is gradual with cycling, and generally several hundred thousand or a few million cycles

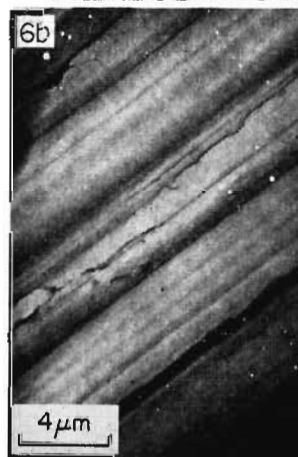
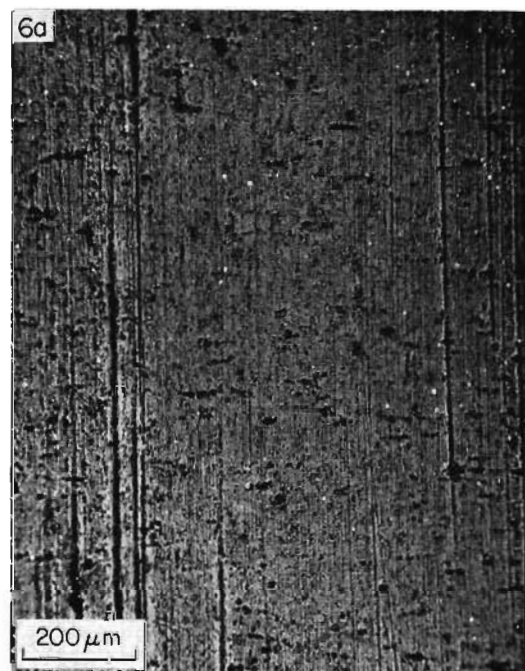


FIG. 6. Surface distress. (a) Optical micrograph of moderate surface distress. (b) Replica electron micrograph of unrun surface. (c) Replica electron micrograph of surface after running.

are required for surface distress first to appear. Surface distress can lead to functional failure of the rolling elements by two mechanisms: first, if overly severe, it appears to lead to superficial material removal such as would occur in wear, leaving behind a rough and "rutted" surface. More often, the presence of surface distress can lead to premature spalling fatigue (8).

It is useful to distinguish the origin of smearing (metal transfer) failure from that of surface distress (fatigue). Smearing appears to require a high degree of sliding in the contact. Surface fatigue occurs in practically pure rolling, and in fact, does not seem to occur when high sliding is present. This distinction

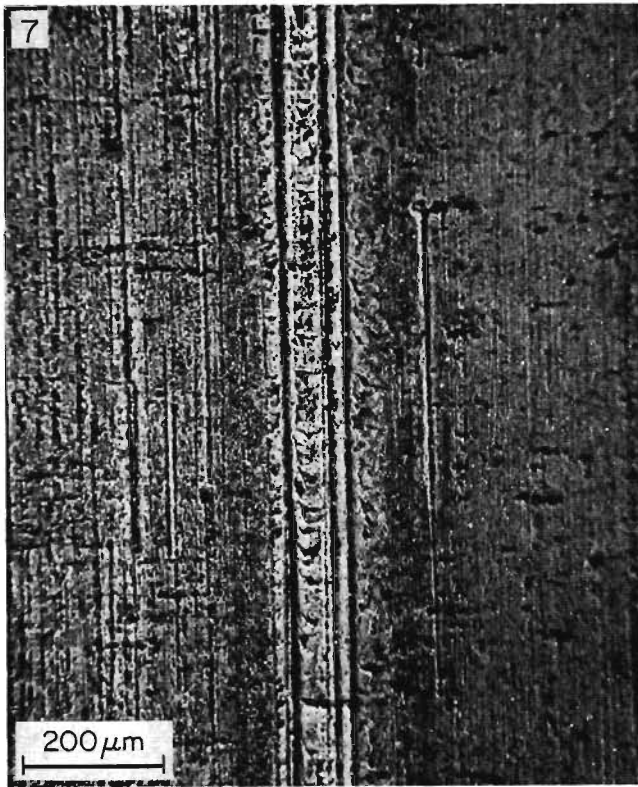


FIG. 7. Optical micrograph of severe surface distress.

will serve as a clue to the mechanism of these failure modes.

This completes the review of contact failure modes of interest to this paper. Bulk failures (e.g., cracking, fretting, and corrosion failures) will not be studied since they are not typically rolling contact failures.

The next section is a review of parameters which are influential in causing rolling contact failure.

II. Parameters influential in causing rolling contact failure

As in all discussions on the strength of materials, it is convenient to distinguish among the parameters relevant to rolling contact failure, a group of stress parameters, and one of strength parameters, the latter term being applied to the solid (generally a hard metal) from which the rolling elements are made.

A failure is expected when the stresses ("severity parameters" of the operating environment) are too great by comparison with the strength parameters.

A Hertzian contact subjects the surfaces and the subsurface material volumes of both rolling elements to a field of elastic stress. In addition, if the contact moves, a temperature field surrounds the contact area. Heat sources are provided by frictional and viscous losses at the contact surfaces, by irreversible deformation in the underlying material, and by external heat inputs to the rolling elements. Heat removal

occurs by all the usual means: conduction through the rolling elements, convection in the lubricant and in the atmosphere, and radiation.

It is convenient to attempt a definition of the complex contact stress and temperature field in steps. First, there are the "bulk" stresses and temperatures existing in the rolling elements at points remote from the contact area. It is generally possible to neglect these as small, except that an ambient temperature of the contact is defined by the surface temperature of the rolling elements at points remote from the contact area.

In a second step, the contact is considered to be between ideally elastic bodies with surfaces (prior to contact) shaped approximately as second-order surfaces. The surfaces are considered geometrically perfect and ideally smooth. There are, then, only normal forces transmitted through the contact which can be considered stationary. This assumption yields the classical Hertz stress field at the contact surfaces and in the underlying material.

Next, one may assume that tractive forces related in some simple manner to the normal forces are transmitted through the contact, and thus arrive again at surface and subsurface stress distributions.

Interposition between the surfaces of an elastohydrodynamic lubricant film modifies the normal surface pressures and thereby the subsurface stresses. The traction forces between the surfaces are also modified if it is assumed that they are transmitted through viscous traction in an elastohydrodynamic film.

The next step is to substitute elastic-plastic bodies for the ideally elastic bodies of the rolling elements and to calculate stresses and plastic flow from this assumption.

Then, real surfaces with finite roughness are substituted for the ideally smooth surfaces with the following effects:

a. Lubrication is no longer necessarily either full film or boundary. There is now a partial elastohydrodynamic regime wherein films form over some portions of the contact area, but asperity peaks pierce the film and come into contact with each other with or without the presence of boundary films.

b. Microstress fields surround contacting asperities and modify the macroscopic stress field of the Hertzian contact area.

c. Tractions are no longer either viscous or Coulomb; mixed traction regimes exist where asperities supply Coulomb traction and the elastohydrodynamic film contributes a viscous component.

d. Junctions may form and wear and metal transfer phenomena may take place between the asperities.

Finally, the ideally homogeneous material of the contacting bodies is replaced by the complex structures existing in real metals. Mechanistically, this will have these main effects:

TABLE 2
Severity Parameters of Contact Conditions

Model	External conditions ^a	Contact parameters ^a	Severity parameters ^a	
			Geometric, kinematic	Stress
Hertz contact	Normal force	Curvatures Elastic constants	Contact ellipse axes Contact deflection (elastic) Accumulated cycles (number of)	Maximum Hertz pressure Maximum shear stress (unidirectional) Maximum shear stress amplitude
With surface traction	Tangential force, moment Sliding speed	Friction coefficient	As in Hertz contact Plus: Slip areas Slide-roll ratio Contact temperature	Shear stress maxima (surface subsurface) Shear stress amplitude maxima (surface subsurface)
With plasticity		Plastic flow stress	As in Hertz contact Plus: Plastic strain Plastic heating	Maximum surface pressure Maximum shear stress Maximum shear stress amplitude Residual stresses
With elastohydrodynamic film	Lubricant quantity Inlet temperature	Viscosity, pressure- and temperature-viscosity coefficient	Dry contact ellipse axes Velocity-viscosity parameter Stress distribution parameter Film thickness parameter Slide-roll ratio Accumulated cycles	Maximum surface pressure Maximum shear stress (unidirectional) Maximum shear stress amplitude Friction force (moment) Maximum contact temperature
With rough surfaces	RMS roughnesses Asperity spacing Asperity slopes		As for EHD film, Plus: Film thickness/roughness ratio Slide/roll ratio Track length covered Accumulated cycles	As for EHD film plus: Microstresses near surface Microtraction Junction growth measure
Strength criteria				Fatigue life Plastic flow strength Surface fatigue criterion Smearing criterion Wear constant

^a The conditions listed for each refined model are additional to those given for the simpler models above.

a. Inhomogeneities will act as local stress raisers.

b. The crystalline anisotropy of the material will cause localized slip during cyclic stressing, which in turn induces residual microstresses.

c. Residual macrostresses may pre-exist in the contacting bodies prior to cycling.

In any of the contacts, heat develops wherever friction occurs (there is also heat developed by plastic deformation in the contact material), and temperature fields accompany and add thermal stresses to the elastic stress fields.

Table 2 illustrates how each external variable enters the picture and modifies the severity parameters as the model of the contact is refined.

When *rough surfaces* in the model are considered, one needs at least three geometrical measures of the roughness: the rms roughness, the typical spacing between the asperities, and a typical asperity slope. Of these three, two are independent.

To round out Table 2, a listing of "strength conditions" is given at the bottom. According to the standard method of strength of materials analysis,

failure prediction is based on a comparison between an appropriate "severity parameter" of stress and the applicable strength condition. As an example, in uniaxial tension of ductile metal, the 45° shear stress is compared with the yield strength for shear, and plastic failure is predicted if the former is larger.

For some failure modes (e.g., smearing), neither the severity parameters nor the corresponding strength conditions are precisely known, and it is therefore difficult to separate the two clearly in Table 2. In this respect, the table outlines a program of inquiry rather than an available method.

To fix ideas regarding the specific nature of the severity parameters, the following brief discussion is presented:

The severity parameters that directly determine the failure are mostly elastic stresses and their location parameters with respect to the contact area. The maximum Hertz pressure in the contact is the most basic of all stress parameters, although no failure mode has been specifically attributed to it. There are two relevant shear stresses. The best known is the maximum *unidirectional* shear stress, i.e., the highest value of shear stress arising anywhere in the contact. (In the absence of traction, this stress occurs at a fixed level below the surface.) This shear stress appears responsible for many of the plastic phenomena observed in contacts.

The maximum *alternating* shear stress is the greatest total amplitude of shear stress to which any plane at any point in the contact is subjected as the contact passes over it. In the absence of high traction, this stress is below the surface at a depth less than that of the maximum unidirectional shear stress. It is held responsible for spalling fatigue. The magnitude and location of all subsurface shear stresses are modified by plasticity, EHD films, and, of course, by any traction imposed on the contact.

In the presence of traction, or when asperities of the surfaces meet (even without external traction), there are high stresses in the surface (macroscopic or localized at asperities), and these surface shear stresses are believed relevant to surface fatigue (surface distress) failure and to wear.

Among the severity parameters which are not elastic stresses, one finds cumulative plastic strain which determines the magnitude of the plastic deformation and the redistribution of stresses due to plasticity, and the elastohydrodynamic "uniform" film thickness which prevails over most of the contact area when loads are high, and, in combination with the parameters of surface roughness, is believed relevant to surface fatigue and wear. There are elevated local temperatures in the contact (due to viscous losses, Coulomb losses at asperities, plastic losses in the contact material), and the highest of these is relevant to smearing failures and to plastic failures of the temperature imbalance type.

III. Analysis of failure modes

After surveying the severity parameters relevant to rolling contact, the typical rolling contact failure modes will now be discussed in more detail.

1. WEAR FAILURES

A. Mild wear by loose particle removal. In fluid-lubricated rolling bearings, operating without solid contaminants, the wear experienced in the absence of considerable sliding is generally mild. It has been found (9) that under these conditions, wear takes place by loose particle removal. Using a radio-activated steel element, which rolls against inactive steel, and performing autoradiography on the inactive elements, it is found that strongly held transfer particles occur only if the lubrication is by a small quantity of oil such that there is an opportunity for loose wear particles to be re-deposited and subsequently rolled into the surface. If there is copious lubrication which will remove loose particles before they are rolled over, then no strongly held transfer particles are found.

Accordingly, the wear phenomenon can be described by giving the wear rate as a function of operating conditions. In (9) and (10), experiments are cited in which the wear rate was determined for a number of mineral oils and synthetic fluids used to lubricate balls rolling on each other in the configuration illustrated in Fig. 8. The wear rate of only one (acti-

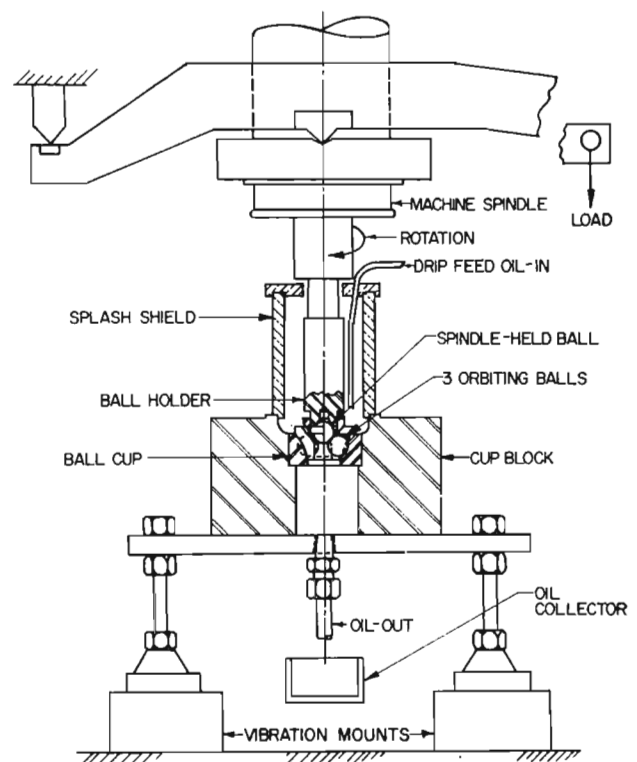


FIG. 8. Rolling four-ball configuration.

vated) ball was measured by scintillation counting of the lubricant and of washings from the test configuration. A run-in period exists if the test is started with new balls, and thereafter the wear rate becomes steady. The experiments were conducted in the partial elastohydrodynamic range, and in this range, the equilibrium wear rate can be expressed as a function of the following variables: (a) the real area of asperity contact, and (b) the slide/roll velocity ratio.

For constant geometric conditions, i.e., constant slide/roll velocity ratio and constant contact material, the wear rate can be expressed to good approximation as follows:

$$w = Ka = Ka_0(a/a_0), \quad [1]$$

where K is a rolling wear constant dependent on the lubricant and a is the area of real asperity contact. This area is obtainable by multiplying the relative magnitude of the real contact area a/a_0 (a function of the film thickness/roughness ratio ξ_0) and the area of the Hertz contact a_0 . The ratio ξ_0 is defined as

$$\xi_0 = \frac{h_0}{(\sigma_1^2 + \sigma_2^2)^{1/2}}, \quad [2]$$

where h_0 is the uniform EHD film thickness in the contact area, and σ_1 and σ_2 is the rms roughness of the contacting surfaces.

Figure 9 shows plots of the wear rate as a function of the film thickness-roughness ratio ξ_0 for a number of lubricants. (Most of the tests were run at a load corresponding to 680 kpsi maximum Hertz pressure.) It is seen that the plots are generally straight lines with somewhat varying slope. If K was a true constant, the slope would be that indicated by the dotted line, representing a/a_0 . Deviations from constancy appear to be related to changes in the boundary lubrication behavior of individual lubricants as a function of the severity of the contact condition (generally boundary lubrication in these tests became better as the asperity contact area became larger (reduced ξ_0), due, perhaps, to chemical reactions in the lubricant brought about by exposure to "fresh" metal surfaces at the points where junctions have been broken).

It is well to bear in mind the magnitude of the wear rates shown in Fig. 9. The highest rate on the plot is 10^{-4} mg/inch track length, and it is observed for $\xi_0 \approx 1$. This amounts to removal of a layer of approximately 400 μ inches thickness from the rolling track during the time the balls travel one million inches on each other. This is quite discernible wear for a rolling contact, and in long running will destroy the accuracy of the configuration. Thus, a contact operating under the very high Hertz pressure of 680 kpsi cannot run long with the thin elastohydrodynamic film corresponding to $\xi_0 = 1$. For higher values of $\xi_0 \approx 3$, the wear rate is about 100 times

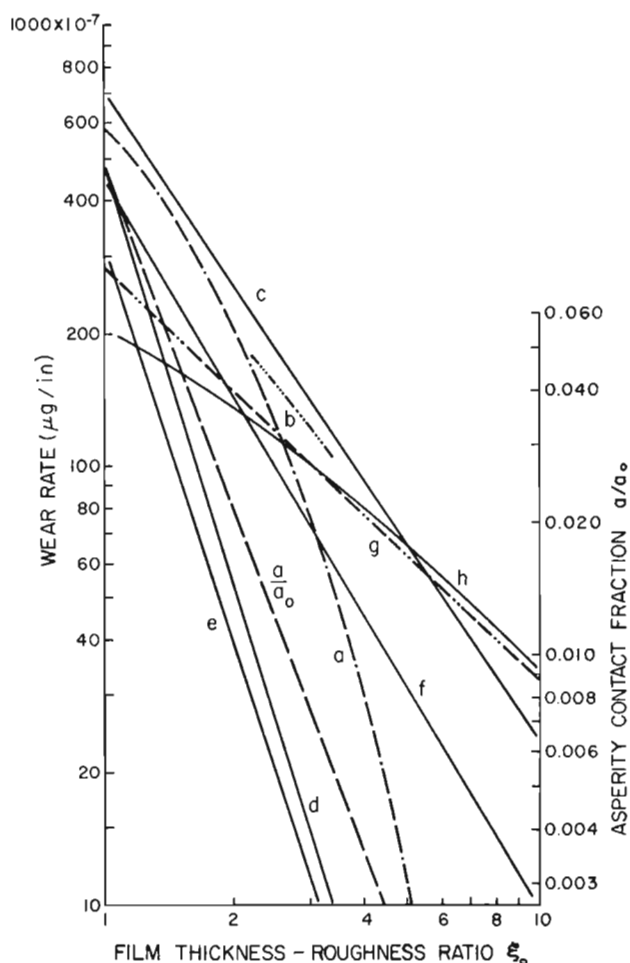


FIG. 9. Wear rate per unit track length as a function of the film thickness/roughness ratio. (a) Mineral oil, 50-lb load. (b) Mineral oil, rough ball. (c) Mineral oil, 150- to 182-lb load. (d) Diester. (e) Ester. (f) Polyphenylether. (g) Mineral oil-decaline, 15.4 cs. (h) Sodium-base bearing grease.

lower, and at these levels it is negligible as a failure mode.

The wear law cited states that the wear constant K is (as a first approximation) independent of the elastohydrodynamic condition, and characterizes wear properties (boundary lubrication properties) of the lubricant-material combination of the rolling contact for the specific geometry and kinematic situation of a test.

The effect of the magnitude of sliding on wear rates seems to depend on the elastohydrodynamic condition. With an almost complete elastohydrodynamic film, the effect of sliding on wear seems to be minor. When there is less film and more substantial asperity contact, the wear rate increases by a factor of ten and more as a ball-to-ball configuration is made to roll with a high amount of sliding (typical of high contact angle ball bearings) instead of zero sliding.

One can use the described concepts of mild wear

of well-lubricated rolling contacts to compare lubricants for their wear preventing action in rolling contact, and to predict the wear life of a practical configuration on the basis of relatively simple wear experiments, providing that the lubricant, surface geometry, and contact configuration are identical in experiment and practice, and that ξ_0 is known in both cases.

Wear is a serious problem in rolling bearings lubricated with solid lubricants or very poor liquid lubricants such as water, liquid metal, or cryogenic fluids. There is a considerable volume of practical test data on hand (11, 12) reporting on wear behavior of solid lubricated bearings in specific applications, but generalizations have not been published that would permit prediction of wear behavior. The same can be said for bearings operating in an environment leading to contamination by abrasives, e.g., sand. The wear process in the presence of sand still appears to be "mild," i.e., there is usually no smearing, but the wear rate can be very high and lead to early failure of the contact due to loss of dimensional accuracy.

B. Severe wear (smearing). Smearing is a serious failure of Hertz contacts that is very common in gears and is also encountered in rolling bearings, primarily at surfaces operating with a high degree of sliding (e.g., roller end/flange contacts). It is also a serious problem in bearings subjected to high angular accelerations or running at extreme speed and low load where, due to the low traction transmitted through rollers or balls, they may stop rotation around their own axis and subsequently undergo sliding. The typical smearing failure illustrated in Fig. 2 varies considerably in appearance due to geometrical differences between contacts, but it is always characterized by substantial transfer of metal from one surface area to another, on the same or the opposing surface. The time required to form a smearing failure, if conditions are conducive to smearing, is extremely short. Macroscopically visible metal transfer can occur within inches of sliding after the moment when the onset of severe wear is first indicated by an increase in traction. Figure 10 shows the wear scar on a hard (750 DPH) steel ball which was slid over another ball. The transfer particles are colored deep black as a result of an autoradiographic technique, and are deposited in rows in the direction of sliding. If this type of failure is allowed to progress, it is generally self-aggravating and leads to rapid destruction of the contact.

The rate of deterioration of a smeared junction depends greatly on the conditions of contact. Cocks (13) has described a wedge-formation process by which the generation of transfer can be explained. This mechanism suggests that the severity of transfer can be reduced if asperity junctions are separated after a short sliding distance. The observation of Hertz contacts undergoing smearing failure reported

in (14) and (15) substantiates the belief that the damage is more rapid if junctions are allowed to grow, and less rapid if they are positively separated after a short distance, even if the same surface elements undergo repeated sliding contact.

Flash temperature calculations as introduced by Blok (16) can be used to yield an order of magnitude severity criterion for the onset of smearing. By calculating the maximum flash temperature in a contact, it is possible to judge whether it can attain a magnitude at which a transition could be expected for a given lubricant [see, e.g., (17) for methods of calculation]. Transition temperatures for many

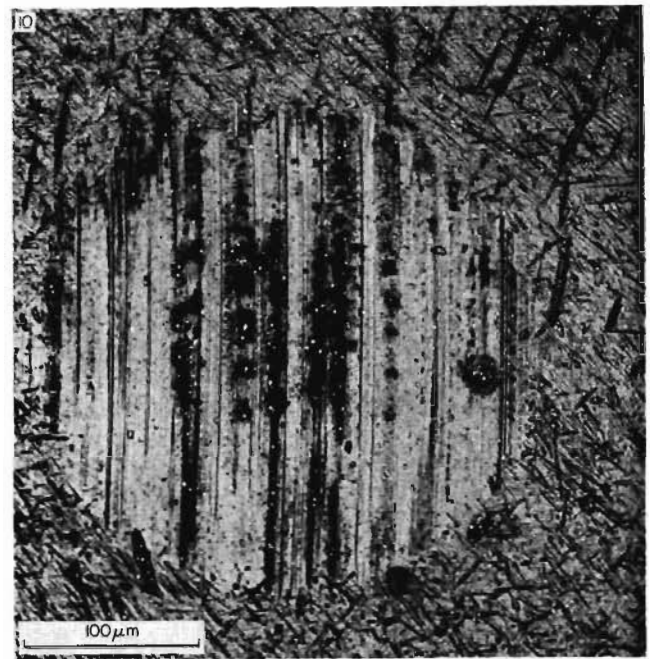


FIG. 10. Wear and smearing (metal transfer) on a steel ball sliding over another ball. Wear scar, 50-kg load.

straight mineral oil lubricants are of the order of 150 C, according to Kelley (18). For lubricants with so-called extreme pressure additives, the transition temperature is higher.

If it is desired to refine the prediction of the severity parameters of a smearing failure beyond an order of magnitude estimate, then it is necessary to conduct direct experiments under conditions simulating the intended application. These lead to engineering rules for the calculation of smearing limits for a class of contacts. The application of such empirical methods to rolling bearings is complicated by the fact that the contact forces or sliding velocities under which smearing occurs in a rolling bearing may be unknown because they are internally generated. (The magnitude of sliding is unknown for a ball or roller in a bearing under high acceleration. Forces are unknown between roller end and flange in a radially loaded cylindrical roller bearing.) The

problem must therefore be tackled in two steps: first the contact load conditions are ascertained by methods of experimental stress analysis, and second, a transition criterion, applicable to the materials and operating conditions, must be found. One can conduct sliding or rolling-sliding experiments with well-defined contact geometries simulating those of rolling bearings and using known speeds and loads, e.g., in a contact between two hardened steel balls. In (14) and (19) experiments are described in which a hardened steel ball was made to slide on an identical stationary ball under loads comparable to those encountered in the contacts of rolling bearings. Measurements of load and friction torque were made, and contact temperature was extrapolated from thermocouple measurements in the vicinity of the contact area. Each test was performed at constant speed and increasing load, and a 10:1 speed range was covered for a variety of lubricants. The transition from mild to severe wear (smearing) was noted in each case by using the torque rise that accompanies the onset of smearing. Results indicate that for a given lubricant and constant ambient temperature, the load and speed at the transition point are related as follows:

$$W \sim S^{-\epsilon}, \quad [3]$$

where W = contact load, S = relative speed of the surfaces (sliding speed), and $\epsilon \approx 0.5$.

The transition load at constant speed varies, of course, with the lubricant. The results show a well-defined and repeatable transition to smearing for boundary-lubricated hard steel Hertz contact surfaces running with a high degree of sliding. A similar transition to severe wear has *not* been observed for pure rolling or little sliding, i.e., it appears that (at least at ambient temperature in atmospheric air) hard steel Hertz contacts can be rolled on each other with almost any lubrication or unlubricated, and at loads up to the plastic range, without typical smearing failure. Presumably, under pure rolling, the transition criterion is not exceeded because of the low energy dissipation in the contact. Of course, the criterion can be exceeded, even in rolling, in hard vacuum.

Boundary lubrication by laminar solid coatings or by the oxides of the contact material can effectively prevent smearing, and the difference between dry-lubricated and liquid boundary-lubricated contact operation appears to be a matter of the specific load that can be carried. The transition criterion for dry-lubricated surfaces is likely to be ill-defined, however, because it cannot be taken for granted that the dry lubricant is always available to replenish an existing boundary film in the contact.

It should be noted that in the presence of high degrees of sliding, elastohydrodynamic film thickness calculations are precarious because heat dissipation in the contact drastically changes lubricant viscosity.

It has been observed (19) that elastohydrodynamic films become unstable and break down even at high rolling speeds when sliding is increased. Therefore, smearing failure can occur when sliding is high in spite of the fact that the rolling speed component is sufficient to yield a substantial isothermal elastohydrodynamic film thickness.

2. FATIGUE FAILURES AND PLASTIC PHENOMENA

In the classical literature on rolling bearings (2), plastic deformation and spalling fatigue failure are treated as two distinct phenomena. The former is attributed to overload of the contact to a point where directly measurable plastic depression of the surfaces occurs; the latter is a result of prolonged cycling at lower loads. From the point of view of failure analysis, the two regimes are not clearly distinct. In (20), it is shown that there is interaction between spalling fatigue failure and plastic flow in ball contacts. In (21), plasticity occurrences on a microscopic scale are identified in normal bearing fatigue test situations where measurable plastic surface deformation does not take place.

In spite of this, it is clear that primary failure by cumulative plastic flow is an overload phenomenon. The stress level at which failure occurs depends on the permissible limit of plastic deformation. A definition based on permissible plastic diameter reduction equal to 10^{-4} times the rolling element diameter is given in (2), and a static load rating method of bearings has been based on it. As an order of magnitude statement for hard (>700 DPH) steel as used in rolling bearings, one can expect noticeable bulk plastic flow to take place if the maximum calculated Hertz pressure exceeds 500 ksi.

Lubrication influences gross plastic flow in a manner which is not yet well understood. The tracings across the rolling track on a ball shown in Figs. 1 and 3 represent test runs made under identical load, but at two different speeds (Fig. 1 at 2500 rpm and Fig. 3 at 14,500 rpm). The deep depression in the center of the track in Fig. 3 indicates that apparently more plastic flow has taken place at high speed ($\xi_0 = 5.2$) than at low speed ($\xi_0 = 1.7$). The generality of this result has not yet been demonstrated.

It was mentioned earlier that a type of plastic flow failure takes place when a rolling contact reaches an unstable thermal condition where less heat is dissipated than produced so that the contact overheats, the lubricant is destroyed, and subsequently the contact material is softened. These phenomena differ from the "cold" plastic flow failures observed in temperature equilibrium only in degree.

To illustrate the effect of bulk plasticity on fatigue life, it is convenient to examine the load-life law of rolling contact. In the nonplastic region, a group of identical rolling bearings have a statistical life [L_{10} life as defined in (2)] inversely proportional to

TABLE 3
Stress-Fatigue Life Relation of Point Contacts

Regime	Test specimens			Stress life relations
	816 DPH balls on 720 DPH flats	816 DPH balls on 772 DPH flats	4-ball test, 800-832 DPH balls	
	Limit stress (ksi)			
Plastic, high stress	>880 ~ 640	>750	>800	Approximately inverse 9th power
Plastic, medium stress	640 ~ 500	750 ~ 550	800 ~ 750	No substantial increase in life for reduced stress.
Plastic, low stress	<500	550 ~ (<500)	750 ~ 600	Life increases faster than inverse 9th power for reduced stress.
Elastic, service stress	<<500	<<500	<<600	Inverse 9th power

the third or fourth power of the applied load (depending on contact geometry). This load-life law is violated in the plastic range as described in (20) and illustrated in Table 3. Apparently, depending on hardness and stress level, there are severe deviations from the predicted load-life law, which can be interpreted (20) as interaction phenomena between plastic flow and spalling fatigue. As a result, fatigue life in the plastic range depends greatly on configuration and absolute stress level.

In the true fatigue failure regime, i.e., at stresses where macroscopic plastic flow does not take place, the statistical spalling fatigue life of a group of rolling contacts follows the general formula propounded by Lundberg and Palmgren (22) and since then extensively standardized. This formula can be written as a Weibull distribution of lives to failure:

$$P \sim f[(\tau_0^e/Z_0^h)N]^e V, \quad [4]$$

where P is the (cumulative) probability of failure within cycles, which is shown to be a function of the maximum alternating shear stress τ_0 , its depth Z_0 , the number of stress cycles N , and the stressed volume V . (This latter is never defined in absolute units; it merely serves as a size factor in scaling experimental results.) The exponent e (Weibull slope) is a dispersion parameter, and it is generally of the order of 1-1.5. The exponents e and h for stress and location are of the order of 10 and 2, respectively, although the precise values differ depending on whether point or line contact is involved, and there is some controversy over them. f is an exponential type function.

In discussing the true fatigue failure regime, we will consider rolling contact conditions with only minor sliding so that severe wear (smearing) does not take place, lubrication that keeps the mild wear rate negligible, and an adequately maintained temperature balance. Unless extraneously caused failures such as bulk cracking occur, only fatigue failure modes remain to limit the life of such a

contact. As described in Section I, there are two fatigue failure modes: surface fatigue and (sub-surface) spalling fatigue. These two failure modes can be considered as extremes. Pure spalling fatigue without any surface fatigue phenomenon signifies extremely good lubrication and smooth surfaces (or, alternately, very potent subsurface stress raisers such as large inclusions). Predominant surface fatigue (often followed by a premature onset of spalling) signifies inferior lubrication, i.e., that the elastohydrodynamic and/or boundary lubrication parameters are inadequate for the operating conditions and for the surface characteristics selected.

The criterion that determines which of the two modes of fatigue failure predominates appears to be closely related to the elastohydrodynamic film thickness/roughness ratio ξ_0 defined in Eq. [2].

To explore this dependence, the life to *spalling* fatigue failure will first be discussed as a function of ξ_0 .

If one plots statistical life (L_{10}) of many homogeneous groups of rolling elements operated at varying ξ_0 values, but at identical load, a curve of the type shown in Fig. 11 results. The numerical values on the abscissa and ordinate vary, depending on the contact configuration, the chemistry of the lubricant, and the material of the contacts. In all cases, however, the curve has a "knee" as shown in Fig. 11 around $\xi_0 \approx 1$. Above this point, at least up to $\xi_0 = 4$, life increases gradually with speed as reported in (20, 23). Life increase is more pronounced for overloaded contacts in the plastic regime due to the deviation of such contacts from the third power load-life law (20), but considerable (2- to 10-fold) increases have also been observed in full scale rolling bearing tests where there is no bulk plastic flow (23).

The range above $\xi_0 = 4$ is less well explored, but there is evidence from extremely high-speed operation of gas turbine ball bearings that further life increases may be available in the fully elastohydrodynamic

region. This could be attributed to the fact, shown by elastohydrodynamic theory, that the alternating shear stress is reduced at high speeds by elastohydrodynamic pressure redistribution.

At some value below $\xi_0 \approx 1$ (depending on lubricant chemistry and probably on the contact material), there is an abrupt drop in spalling fatigue life as reported in (8), and similar behavior has been observed in many other tests (on through hardened ball bearing steel) in the author's laboratory. This abrupt drop in spalling fatigue life (by a factor of 40 or more) is accompanied by the visible onset of surface distress as illustrated in Figs. 6 and 7. In fact, the reduction of ξ_0 below a critical value appears to

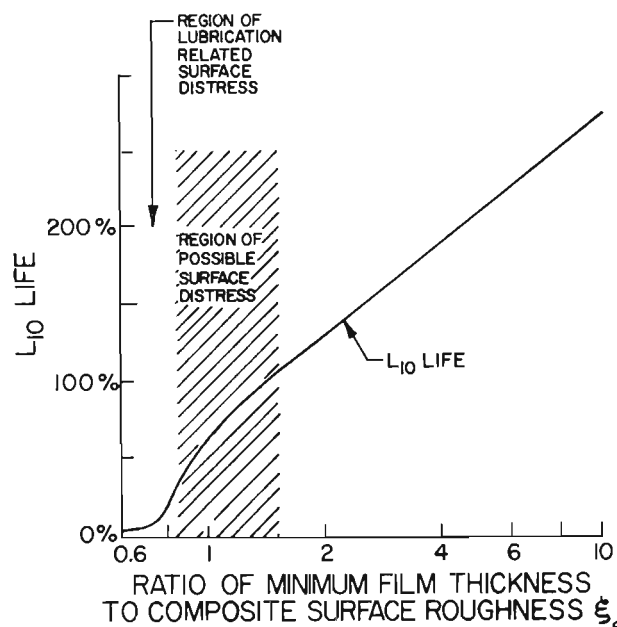


FIG. 11. Group fatigue life L_{10} as a function of the film thickness/roughness ratio ξ_0 .

result directly in surface distress and only indirectly in a drop in spalling fatigue life, because visible signs of surface distress are virtually always observed for these low ξ_0 values, but early spalling failure does not occur in all tests.

The severity of surface distress depends by no means solely on the value of ξ_0 . The boundary-lubricating abilities of the lubricant and the nature of the rolling surfaces (e.g., solid lubricant coatings) can retard or promote surface fatigue. However, it appears that by maintaining ξ_0 values in excess of $\xi_0 = 1.5$, generalized surface distress is avoided [assuming, of course, that the ξ_0 value is accurately known (measured) under the test conditions].

New understanding regarding the nature of rolling contact fatigue failures (both spalling and surface fatigue) has recently been obtained by metallurgical study. In (24), Bush *et al.* have described electron microscopic observations of structural changes beneath

the rolling surface in a cycled contact. In (25), Reichenbach and Syniuta give electron microscopic observations of surface changes under operating conditions considered by the present author to be conducive to surface fatigue.

A coherent description of both fatigue mechanisms is currently being attempted in the author's laboratory. Results to date have been published by Martin (6, 21). From today's vantage point, the sequence of events leading to a fatigue failure in a hard steel rolling contact is the same for both spalling and surface fatigue and is roughly as follows:

In zones of high shear stress, small confined volumes of material undergo a gradual transformation with cyclic stressing. If the transformation has proceeded far enough, it becomes visible under the microscope if a section is made through the material under the contact and suitably etched. This is shown in Fig. 12 in which a section containing white etching-transformed structures is overlaid with contours of equal shear stress level. At high magnification in a section taken along the rolling track, the transformed white etching material appears as a series of narrow slanted white lines accompanied by black oblong objects as shown in Fig. 13. By taking sections in several planes oriented at different angles, it has been shown that the transformed zones form thin two-dimensional lenticles. By observation of the rolling direction of the ball, one sees that the white lines form in a direction rising toward the rolling surface in the direction of ball travel. The angle included with the surface is typical of the configuration (23° for a deep groove ball bearing).

Under the transmission electron microscope, the white etching areas compare with the unstressed regions of the same material as shown in the two halves of Fig. 14. Electron diffraction analysis has indicated that the small circular cells in the altered material exhibit a high degree of crystal fragmentation as found in highly plastically deformed material. The cell walls are dislocation tangles. It has also been shown that the black elongated structures accompanying the altered matrix are carbides which formed during the cycling. On the other hand, temper carbides seem to disappear from the altered structures. Thus it appears that as a result of prolonged cyclic stressing, there are microplastic deformations taking place in the high shear regions which substantially alter the material. Observation has also shown (26) that plastically deformed structures surround non-metallic inclusions located at the depth of high shear stresses (Fig. 15).

A topographic analysis of fatigue spalls reveals that the spall bottom consists of: (a) a series of fracture planes running downward at an angle from the rolling surface until they hit one of the plastically deformed bands, and (b) planes running along the boundary of the deformation bands toward the rolling surface.

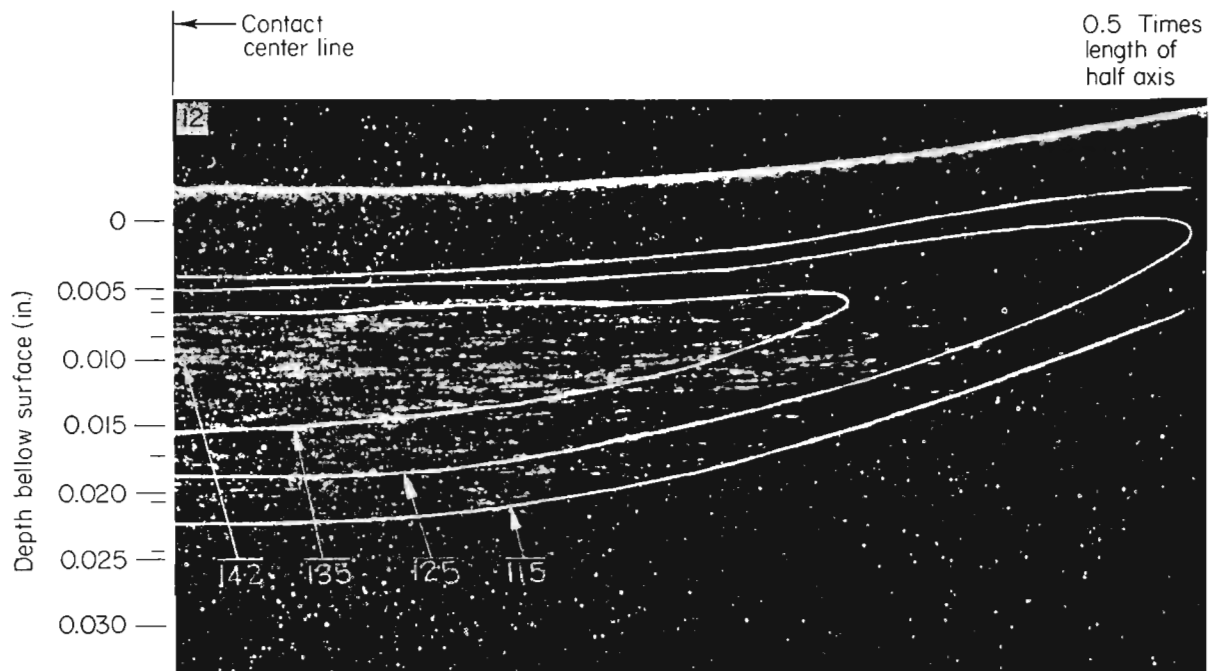


FIG. 12. Section through a ball bearing ring showing white etched deformation bands in the high shear stress region (numbers on the photograph are shear stress levels, ksi).

Recent observations (6) have also revealed that plastic deformation bands can be found very close to the rolling surface at locations which have undergone surface fatigue. This is illustrated in Fig. 16, which shows a spall which, at the point marked, runs into a relatively deep scratch mark left over from finishing operations on the rolling surface. By sectioning through the scratch mark, the micrograph shown in Fig. 16b was obtained. Clearly, deformation bands run right up to the surface and into the spall. When observing the rolling surface in the vicinity of a deep scratch, the bright fringe containing many micropits shown in Fig. 17 appears.⁴ This is an indication of surface distress. The significance of this observation lies in the fact that around a deep scratch, the stress field is distorted. First, the elastohydrodynamic film is interrupted at the edges of the scratch by side leakage of oil into the scratch (the depth and width of the scratch are both large by comparison with the elastohydrodynamic film thickness). The absence of film around the scratch can account for the surface distress. Second, the sharp edges of the scratch lead to notch stresses as predicted by Lundberg (27) and recently shown in more detail by Chiu (28). The high traction between surfaces not separated by an elastohydrodynamic film adds to the stresses. In this stress condition, high shear stresses arise close to the surface. This correlates with the appearance of the deformation bands. Of course, a similar explanation ac-

⁴This surface was lightly etched prior to stress cycling to emphasize the bright fringe.

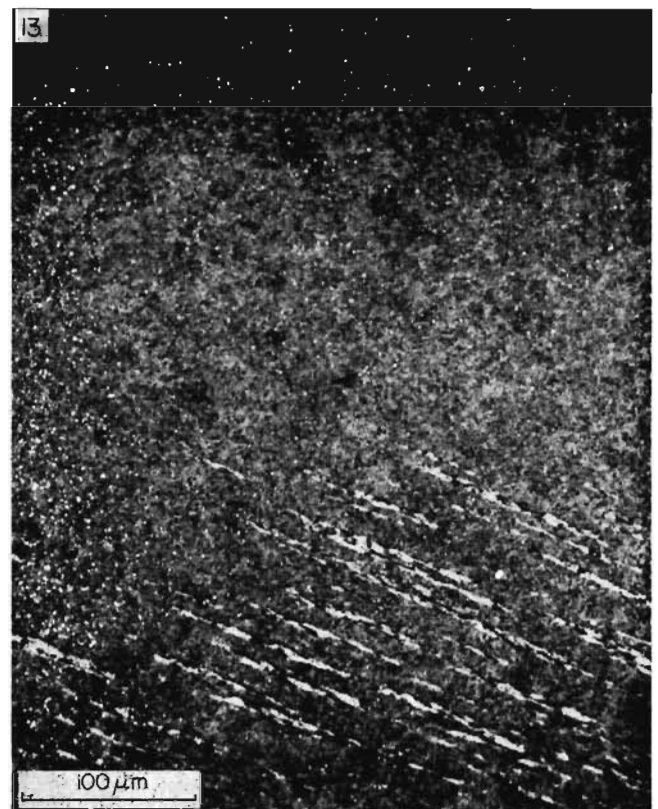


FIG. 13. Section along ball track showing 23° orientation of deformation bands. White deformation bands oriented at 23° to the horizontal. Direction of ball travel is from right to left.

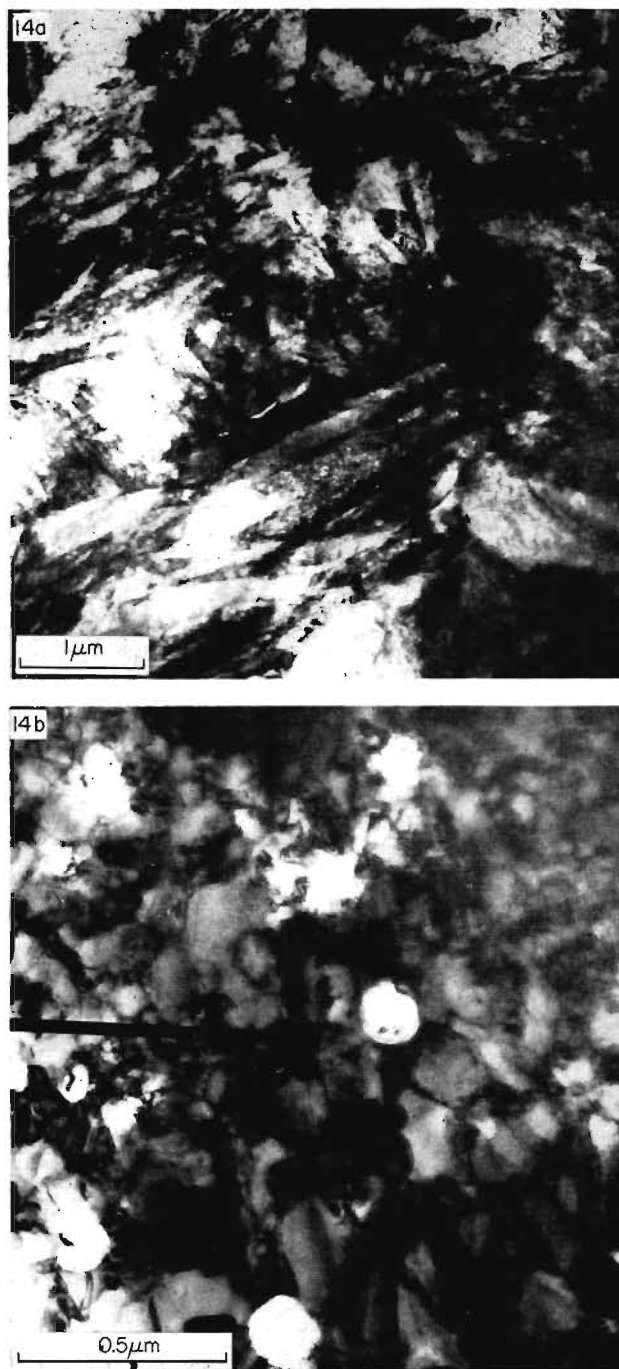


FIG. 14. Stress-induced structural alteration of martensite. (a) Unstressed, tempered martensite. (b) Deformed structure in cycled martensite.

counts for deformation zones surrounding inclusions which are also stress raisers.

It is clear that in rolling contacts operating under loads insufficient to cause bulk plastic flow, there are extensive microplastic occurrences taking place in high shear stress regions. Such occurrences amply suffice to explain the formation of fatigue cracks.



FIG. 15. Deformation bands (white) surrounding a group of nonmetallic inclusions (large black spots).

Indeed, cracks have been observed repeatedly in deformation bands. The geometry of the spalls also suggests that crack propagation is correlated with deformation band positions.

A major problem of rolling contact fatigue, awaiting solution, arises from the observation that in normal material (away from weak volumes) the deformation bands form in the depth range around the maximum *unidirectional* shear stress, whereas life of rolling bearings has been related by extensive statistical evidence to the maximum *alternating* shear stress (22, 29). In fact, the currently used standard fatigue life calculations for rolling bearings are based

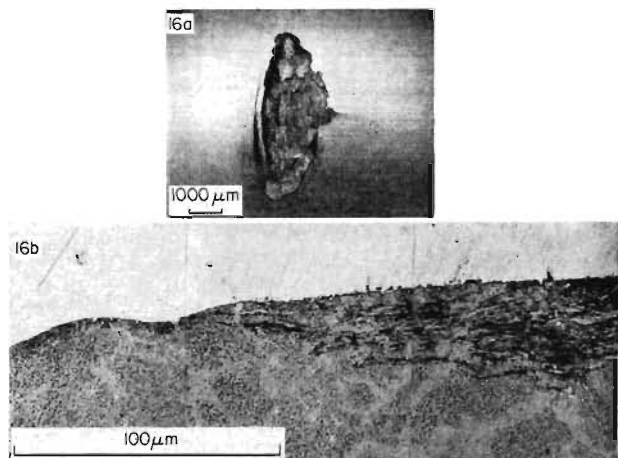


FIG. 16. Fatigue spall associated with surface defect. (a) Spall. (b) Deformation bands leading into a spall.

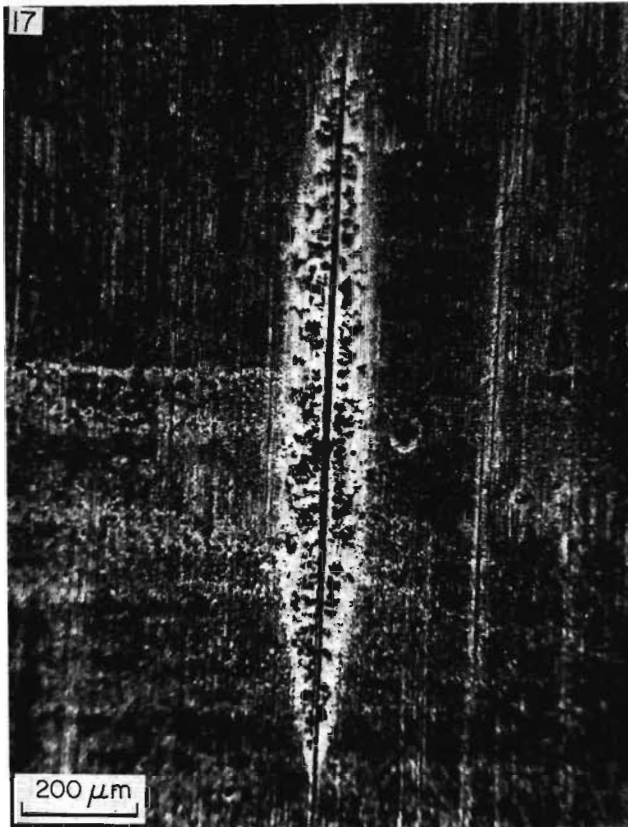


FIG. 17. Burnished surface surrounding scratch.

on this concept and would have to be modified in a major way if the maximum unidirectional shear stress was used as the severity parameter, since the two stresses are not proportional to each other if the contact geometry varies.

The reconciliation of this difficulty will probably come from the recognition that fatigue spalls—in currently available test specimens—arise at weak material volumes (either volume imperfections such as inclusions or surface defects such as scratches). In the absence of plastic flow, the local stress field around these imperfections will be proportional to the undisturbed field at the position of the weakness. Depending on that location within the contact zone, it is conceivable that either or neither of the two macroscopic shear stress maxima governs the stress history of the weak volume. It is also clear that the elastically calculated stresses in the undeformed specimen cease to be relevant to failure as soon as local plastic flow has occurred. For all these reasons, the selection of the most suitable macroscopic severity parameter may depend on the operating conditions (e.g., elastohydrodynamic condition), the type and location of predominant weak volumes, and the overall stress level of the test.

It is also noted that metallurgical evidence available at this time is compatible with the hypothesis that

the microplastic occurrences observed in the material and the nucleation of a fatigue crack leading to spalling fatigue are concurrent but not identical phenomena. The two interact strongly; nonetheless, it is conceivable that time required for initiation or propagation of a fatigue crack to a critical size depends on the magnitude of the maximum alternating shear stress, whereas the growth of deformation bands depends on the maximum unidirectional shear stress.

In the presence of these unresolved questions, one is limited to the following facts:

(a) Current life calculations for spalling failure of rolling bearings, based on the maximum alternating shear stress derived from Hertz's assumptions, supply reliable life information for bearings operating under specified "normal" load, speed, and lubrication conditions.

(b) All phenomena relating to rolling contact fatigue observed so far in hard steel specimens have been shear stress dependent. Accordingly, the microplastic occurrences observed in cycled material are an indication of a danger zone in the material even though they may not pinpoint precisely the area in which, in an undisturbed volume of steel, cracks are most likely to originate.

This information leads to several practical conclusions. First, the occurrence of microplasticity in any unsuspected location indicates a potential source of fatigue failure. This is the case around nonmetallics and, as now known, it is the case in the vicinity of the rolling surface if this surface is subjected to high shear stresses such as around surface discontinuities (scratches) and in the presence of insufficient lubrication leading to generalized surface distress.

Figure 16 clearly shows that the surface distress and formation of deformation bands occur jointly in the vicinity of surface discontinuities even in a well-lubricated contact, and it is reasonable to assume that the generalized surface distress arising in the case of poor lubrication is the same phenomenon. Thus, whenever high shear stresses are generated in, or close to, the rolling surface by a discontinuity or by high traction as a result of poor lubrication and heavy asperity interaction, surface fatigue will result.

The user of rolling contact configurations can exercise control over surface fatigue by proper lubrication and proper selection of the rolling surfaces and of the operating conditions to insure that the elastohydrodynamic film is adequate. On the other hand, the metallurgist attempting to select improved materials for rolling contact applications can study material and treatment variables which influence the reaction of the material to cyclic stresses as evidenced by metallography and endurance test results.

It must be said that the mechanism of flaking failure described here is at variance with the classical findings of Way regarding the influence on pitting

TABLE 4
Summary of Strength Criteria

Failure mode	Strength criterion
Mild wear (loose particle removal)	Wear rate or wear depth per unit track length shall not exceed level permissible in order to obtain required life.
Severe wear (smearing)	Transition to severe wear shall not occur. This criterion is believed to be equal to the following: Flash temperature (modified as necessary for secondary variables) shall not exceed critical level. Smearing occurs only in contacts with substantial sliding.
Cold plastic flow	Maximum unidirectional shear stress shall not exceed plastic flow stress. For arrested plastic flow: maximum shear stress active after plastic flow has stabilized shall not exceed flow stress.
Spalling fatigue	Statistical fatigue life is a function of one of the maximum shear stresses (believed to be maximum <i>alternating</i> shear stress τ_a). This stress shall be kept at level sufficiently low to give required life.
Surface distress (surface fatigue)	Surface distress shall be avoided by maintaining stress condition at and near the surface at permissible severity level (high loads, severe asperity contacts, long cycling cause surface distress if sliding is <i>low</i>). Surface distress appears to be avoided if elastohydrodynamic film thickness/roughness ratio is sufficiently large.
Thermal imbalance failure	Temperature of contact shall stabilize at a value below failure temperature of lubricant and softening temperature of contact material.

failure of oil pressure in a crack (30). Dawson reports current verifications of the Way theory in experiments which appear beyond challenge (31-33). It is believed that the disagreement between the two theories (and their supporting evidence) is due to the often overlooked fact that experiments under conditions simulating gears (as those of both Way and Dawson) are characterized by the use of contact materials of relatively low hardness, and that such materials fail in rolling contact by a different mechanism from the much harder materials used in rolling bearings. This is illustrated by the hardness regime used in Dawson's tests (190-240 DPH) as compared with the hardness of a typical rolling bearing which is always above 700 DPH. To date, no compelling explanation of the substantial difference in failure modes between these hardness ranges has been offered. It is known, however, that the plastic behavior of material in the two ranges is vastly different, and that hardness or hardness-related structural properties of the material are responsible in a major way for the details of crack formation, specifically in rolling contact situations.

3. HEAT IMBALANCE FAILURE

The severity criterion for heat imbalance type plastic failure is the temperature at which heat generated in or transferred into the rolling contact mechanism is balanced by heat removal. This temperature is compared with two strength criteria: the temperature tolerance of the lubricant and the temperature at which the contact material softens. If either is exceeded, heat imbalance failure will result.

4. SUMMARY OF STRENGTH CRITERIA

Table 4 is a summary of the strength criteria discussed with respect to each mode of contact failure. By applying each criterion in turn to a given contact operating in a defined environment, it is possible, in principle, to identify that failure mode which is most likely to cause the mechanism to malfunction. Alternately, one can identify a given failure type by examination of the failed contact and draw conclusions regarding (unknown) operating parameters from the known severity conditions influential in promoting that type of failure. The next step, then, is to engineer the mechanism in such a manner that the severity of its operating environment will provide acceptable life to failure.

Conclusions

There are four major competing types of rolling contact failures: wear, plastic flow, fatigue, and bulk failure. Of these, only the first three arise at the rolling contact. The most important severity parameters of the operating environment have been defined for each failure mode, and this permits a judgment regarding the relative likelihood of ultimate failure by either of the competing failure modes. Engineering action can then be taken to control the most critical failure mode by setting acceptable limits for the relevant operating parameters.

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DISCUSSION

E. V. ZARETSKY AND R. J. PARKER (*National Aeronautics and Space Administration, Cleveland, Ohio*):

While many of the conclusions presented are of a subjective nature, the author does present an interesting paper. The discussors would like to comment upon these conclusions based upon their (the discussors) work and experience at the NASA-Lewis Research Center.

It has been reported (*A1*) that at maximum Hertz stress levels of 600,000, 675,000, and 750,000 psi in the NASA spin rig, stress-life exponents, n [where fatigue life is inversely proportional to stress to the n power, $L \sim (1/S)^n$; or load to $n/3$ power, $L \sim (1/P)^{n/3}$] of 10.4 and 9.7 were obtained for 9/16 inch diameter SAE 52100 balls and M-50 cylinders (races), respectively, with mineral oil as the lubricant. Similar work performed with the General Electric R-C fatigue tester also exhibited a stress life relation exponent of approximately 9 (*A2*). These values were in reasonably good agreement with the values of 9–10 accepted by the bearing industry. As pointed out by the author in his paper, at high stress levels, such as those reported in (*A1*, *A2*), plastic deformation will occur. Apparently, the plastic deformation in the tests of (*A1*, *A2*) did not have a sufficient effect on the contact stress or the material properties of the specimens to affect the stress-life exponent.

Using the author's data, the discussors obtain a stress-life exponent of 13.5 and 11.3 for the L_{10} and L_{50} lives, respectively, or an average value of 12.4 using the method of least squares to analyze the data. These values are much higher than the accepted values discussed previously.

The author attributes plastic deformation of the rolling-element surfaces and metallurgical changes as possible causes of this deviation. The author also indicates that below a threshold value of stress, no gross plastic deformation will occur which concurs with the data reported in (*A3*). Therefore, for purposes of example, assume that no plastic deformation will occur at 600,000 psi maximum Hertz stress and that at 800,000 psi sufficient plastic deformation occurs to cause a reduction in the maximum Hertz stress to 720,000 psi (*A3*). If it is assumed that no plastic deformation occurs, the theoretical ratio of the lives at 600,000 and 800,000 psi would be

$$L_{600,000}/L_{800,000} = (800,000/600,000)^9 \approx 13. \quad [A1]$$

If, however, plastic deformation occurs reducing the stress of 800,000 psi to 720,000 psi, then the theoretical ratio of the lives would be

$$L_{600,000}/L'_{800,000} = (720,000/600,000)^9 \approx 5.2. \quad [A2]$$

If the experimental ratio of the lives was 5.2 as illustrated in Eq. [A2], and if the effect of plastic deformation on stress was not considered in calculating the stress-life exponent, then

$$L_{600,000}/L'_{800,000} = (800,000/600,000)^n \approx 5.2, \quad [A3]$$

and

$$n \approx 6, \quad [A4]$$

which would be the apparent stress-life exponent. It then can be concluded that where gross plastic deformation does occur, its effect would be to reduce the stress-life (load-life) exponent

rather than increase it. Therefore, in the author's paper, since the stress-life (load-life) exponent is much higher than the normally accepted value, gross or bulk plastic deformation cannot account for this anomaly. Additionally, if metallurgical changes at high stress levels affected the stress-life exponent, this effect would be apparent in the experimental stress-life relationships reported in (*A1*, *A2*), which it is not.

Bear and Butler (*A4*, *A5*) used optical microscopy techniques to study half-inch diameter SAE 52100 and AISI M-1 steel balls run in the NASA spin rig at both room temperature (80 F) and at about 250 F, at maximum Hertz stresses from 600,000 to 700,000 psi. Examination of sectioned ball specimens at magnifications up to 1500 $\times$ indicated the same structural alterations with SAE 52100 ball specimens at 250 F as reported by the author. Bear and Butler speculated that the change was to tempered martensite. The same structural change was noted by Bush *et al.* (*A6*). However, Bear and Butler observed no structural change at room temperature. For AISI M-1 steel balls, there were slight structural changes in some of the ball specimens tested at room temperature. At 200 F the change in structure was more pronounced. As in the SAE 52100 steel ball specimens, the altered region became more prominent with increasing running time and expanded in area nearly to the specimen surface.

The author attributes the observed structural alteration in part to plastic deformation occurring beneath the rolling contact surface. However, whether this plastic deformation and the related structural change is due to the effect of maximum unidirectional shearing stress or to maximum orthogonal shearing stress is not really established.

The author bases his conclusion that the critical stress is the maximum unidirectional shearing stress on the fact that the location of the microstructural changes coincides with the location of the calculated maximum shearing stress. It seems equally as significant that the altered regions lie on planes oriented 23° to the surface at the point of maximum shearing stress. Therefore, the critical stress, whether orthogonal or maximum shearing stress or some other stress, is one which has its maximum value at a plane which lies at a 23° angle to the surface at a distance approximately 0.010–0.015 inch below the surface. All of this indicates that the assumption upon which the calculated stresses are based, i.e., nonlubricated, two-dimensional Hertzian loading, is not completely valid for the experimental situation, i.e., full-scale lubricated bearings. Until more exact stress calculations for lubricated full-scale bearings are available, it appears meaningless to speculate as to what stresses are causing the observed structural change. It can be concluded, however, that a structural change occurs in the subsurface zone of maximum resolved shearing stress without any regard to which shearing stress is involved.

It has been suggested in the discussion of (*A6*) that there is a threshold energy input below which the structural transformation is not likely to occur. At higher temperatures, more thermal energy is being put into the test specimen, so that it requires much less input from the cyclic stressing to accomplish the transformation. Martin *et al.* (*A7*) mention that considerable tempering occurs during the hundreds of hours required for testing. Since the microstructural changes apparently result from a thermal-mechanical process, were any tests run at lower temperatures? Was the hardness of the unstressed part of the bearings measured before and after

testing to indicate the amount of tempering occurring during tests?

It was noted by Bear and Butler that, while softening was indicated in the stress-affected zone of the SAE 52100 specimens, a slight hardening was indicated for the AISI M-1 specimens. Additionally, at room temperature no alteration of the structure was noticed in (A4, A5) for SAE 52100 ball specimens, and very little was noticed in AISI M-1; yet there ball specimens had the same failure characteristics as ball specimens tested at about 250 F. The author conducted his tests at a maximum Hertz stress of 475,000 psi and a temperature of approximately 210 F. Fatigue failures in normal bearing operation occur well below these operating conditions. Consequently, if a structural change does not occur at lower stress levels and/or temperatures, it appears rather difficult to relate this change to rolling contact fatigue.

The author presents an interpretation of the transverse profile of a ball track (Fig. 3) which is contrary to the beliefs and experience of the discussors. The area between the straight horizontal line representing the original ball surface and the profile trace below this line (depression) is representative of the amount of deformation and wear that has occurred. The total area of the raised ridges at each side of the track is representative of the amount of deformation. The difference of these two areas is representative of the amount of wear.

In (A8), deformation and wear data was presented for M-1 steel balls in the NASA five-ball fatigue tester at four contact angles, (10°, 20°, 30°, and 40°). Profile traces of the upper ball track were taken after 30,000 stress cycles at 650,000 initial maximum Hertz stress. The data showed that, for the 10° and 20° contact angles, no wear had taken place, i.e., the area above the original surface line equalled the area below the line. However, at 30° and 40° contact angles, progressively greater wear was observed. One would expect wear to be greater at greater contact angles because of the greater sliding (spinning) in the upper-ball (inner-race)/lower-ball contact in the five-ball fatigue tester. In addition, with increased contact angle, the measured deformation area decreased, such that the total deformation plus wear area (depression) stayed essentially constant. This data indicates that when no wear occurs in a rolling/spinning contact at a stress where plastic deformation is present, a transverse profile across the track will show equal areas above and below the original surface line. When the area below the line is greater than that above the line, the difference is representative of wear. Therefore, the profile of Fig. 3 is representative more of wear than of plastic deformation.

The ruts apparent in Fig. 1 are undoubtedly wear, as the author states. But the discussors believe this figure to be indicative of an initial stage of wear. Is it therefore not possible that for the trace in Fig. 3, the wear progressed beyond the "rut" stage to a more uniform removal of material by a wear mechanism across the entire running track?

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- CHARLES H. WEST AND CHARLES A. MOYER (*The Timken Roller Bearing Company, Canton, Ohio*):
- The survey paper Mr. Tallian has provided reminds us of a similar guideline written by F. T. Barwell in 1957 in *Wear* 1, 418-445. Of course, Barwell was concerned primarily with loss of material from contacting surfaces, so his terminology does not exactly coincide and he does not consider plastic flow or all bulk failures in the same light.
- Certainly the present guidelines will encourage re-examination of the great volume of information concerning contact fatigue that has entered the literature since the early work of Stewart Way.
- We note the title of the paper posed by Mr. Tallian and wonder if he means to imply that under a given environmental and bearing condition more than one of the four major failure modes can compete within a given bearing. It seems more likely in a real environment that these major modes may interact and the severe reduction in useful bearing life occurs when two or more contribute to the ultimate loss of serviceability of the bearing.
- For example, under momentary heavy impact loads, localized plastic flow may occur that will jeopardize the elastohydrodynamic film so that surface distress may initiate at the geometric stress concentrations formed by the deformations. Or it may be that metal "smearing" or transfer will take place intermittently under inadequate lubrication conditions and the "pits" formed when the metal is removed can provide sites for fatigue propagation on the bearing raceway surface. Since usually this occurs with low-viscosity lubricant it seems plausible to us that the most likely explanation for propagating the crack is by the hydraulic pressure mechanism.
- This is probably associated with a tangential drag such that spalling seems to propagate more readily on the surface tending to move more slowly through the contact. Barwell cites Milne and Nally (Barwell's reference number 54) as a gross example of this based on the pitting location on the rotating ball of a four-ball rolling ball test rig. Spalls started primarily toward the inner edge of the contact track on the rotating ball whose minor circle moved slower than the mating point on rolling balls. Therefore, the requisites are present to extend a surface crack (or surface defect) subsurface until sufficient

undermining of the raceway occurs, and the usually "arrow-headed" spalling occurs as illustrated in Mr. Tallian's Fig. 16a. We are also curious as to the direction of load application for the contact surface shown in Fig. 5 since the shape of the spall may indicate a surface origin at the left that propagated hydraulically.

Apparently the reason the defect in Fig. 17 and the surface distress in Fig. 7 do not lead to point surface origin failure is that the surface fissures present provide exit paths for the entrapped lubricant so that detrimental hydraulic pressure buildup is unlikely.

The problem of contact fatigue is also somewhat easier to analyze if we recognize the two portions, the initiation and the propagation periods. According to Mr. Tallian's Reference (22), in through-hardened ball bearing material after the initiation, the propagation period was very rapid. In roller bearings, especially with carburized steels, it apparently is a common occurrence to have pits form that, under certain environmental conditions, are nonpropagating. This has forced us to recognize the importance of defining failure and, in fact, the useful life for a bearing in a particular application.

This then brings us to an important point Mr. Tallian alludes to, namely, that the AFBMA and ASA bearing rating standards are based on Mr. Tallian's Eq. [4], which considers only material-associated, subsurface fatigue failures. Whenever the mode of failure deviates from this assumption or if the propagation rate is extended, as is usual for roller bearings, the formula no longer applies so that the bearing rating as defined by Eq. [4] should be looked on as a theoretical value to be used only as an engineering approximation. Certainly the growing understanding of the lubricant and lubricant film as Mr. Tallian outlines is a considerable step forward in helping to remove some of the uncertainties surrounding expectations of antifriction bearing performance in a given application.

AUTHOR'S CLOSURE:

The thoughtful comments of Zaretsky and Parker and West and Moyer are much appreciated. There is no question that both sets of discussors are correct in stating that much material in the paper is of a hypothetical nature, as is inevitable in any attempt to present a system of classification for phenomena as complicated as rolling contact failure. The author feels that it is one of the duties of engineering science to present theories and hypotheses which appear capable of bringing order into diversity of natural phenomena, and thereby permitting a better overview, more systematic interpretation, and, ultimately, better control of the phenomena by the engineer. It is the nature of such theories that they cannot be proved by direct experiment or reference to previous literature as can statements of fact in primary research papers. The theories gradually acquire validity and acceptance as they are compared with an increasing array of experimental observations and appear capable of interpreting and correlating them.

In reflecting in detail on the comments by Zaretsky and Parker, the following may be noted:

The discussors have considerable experience, acquired in their own experiments, with the applicability of rolling contact fatigue life versus load laws. The data they supply are both valuable and interesting. They are used by the discussors to suggest best fitting load-life exponents for the

author's data on high stress circular contact fatigue. However, the point made in the paper is that *no* power law with a single exponent fits the data well throughout the overload region, and that this is indicative of a change in mechanism of the fatigue failure phenomenon compared with that prevailing at lower loads where a power law fits very well.

The discussors also analyze the expected effect of plastic deformation on the load-life law and conclude that plastic flow cannot account for an increase in load-life exponent. This analysis is based on the assumption that plastic flow changes the load-life law essentially by changing the contact geometry. The author continues to believe that metallurgical changes and residual stresses induced by plastic flow can account for almost any type of change in the load-life law, and it is precisely because of the complexity and the many variables inherent in this influence of plastic flow on life that load-life laws given by different experimenters working in the high stress region do not agree.

The discussors mention the concept that the structural changes brought about by rolling contact cycling may, at least in part, be thermally induced, and that the material generated in the process of change is tempered martensite. This is a suggestion first propounded by A. B. Jones a number of years ago. Based on recent information, the deformation bands, which are generated as a result of cyclic stressing, are not tempered martensite. They are a cellular structure of dislocation tangles, generated from martensite by the plastic flow, in which complex carbon migration phenomena take place as described by Borgese (C1).

The author agrees that it is meaningless to speculate whether the maximum unidirectional or the maximum alternating shear stress causes the plastic flow phenomena observed, and that it is probably a stress different from both of those named. However, the author does not believe that it is merely a matter of making more exact stress calculations in order to define a suitable stress. It seems more likely that a better flow criterion will have to be found in order to account for the peculiarities of deformation band formation such as their 23° angle with the surface found in ball bearings. Attempts in this direction have already been made, e.g., by Lyman (C2), and further work is urgently needed.

The question of ambient temperature effect on structural changes is a puzzling one. It would lead too far afield to answer all the detailed questions by the discussors in this space, but it can be said that hardness before and after operation has been determined on the specimens, and that there is evidence that the deformation bands are considerably harder than the unchanged material. There also is evidence that deformation band formation is not directly dependent on the temper resistance of the steel; thus M50 tool steel, which is much more temper resistant than 52100, shows deformation bands being generated at the same ambient temperature and under the same load conditions.

Structural changes in the bulk of the matrix probably do not occur at some of the lower stress levels where fatigue failure is still observed. Martin (C3) has analyzed in detail how a correlation between plastic flow occurrences and spalling failure can still be justified: it is a question of plastic flow in localized areas surrounding weak volumes in the material, where, due to the stress raising effect of these weak elements, plastic flow will occur at substantially lower macroscopic stress levels than are required to precipitate deformation band formation in the bulk of the matrix.

Finally, Zaretsky and Parker suggest that plastic flow can

be deduced from a cross-track tracing of a rolling track only if it can be shown that the amount of material depressed at one location in the track occupies the same cross-sectional area as the amount elevated above the original surface at other points on the track. The author believes that this conclusion is erroneous. It seems to be based on the concept of the incompressibility of plastically deformed steel which would appear to require preservation of volume in the cross section. The reason this concept may be inapplicable is that the plastically flowed volumes in a rolling contact do not rest on a rigid, undeformable substrate. They are, in fact, embedded in an elastic half space, and steel in the elastic range is not incompressible. For this reason, it is entirely conceivable, in fact likely, that the plastic flow induced by rolling a ball over a half space can result in a greater depression in the center area of the track than what elevation of material above the original surface is observed at and just beyond the track edges.

The deficit of volume thus generated can be absorbed by elastic deflection of the bulk material underneath and to either side of the plasticized volume. For this reason, it is more difficult to distinguish wear from plastic deformation based on cross-track tracings than would appear from the concept of balancing cross-sectional areas. The author believes that the only convincing way of doing so is based on the recognition that "ruts" cannot be explained on the basis of plastic flow, and material elevated above the surface cannot be explained by wear. Beyond this qualitative distinction, it seems hazardous to separate the quantity of wear from that of plastic flow based on cross-track tracings only.

Concerning the comments of West and Moyer, the author first wishes to state that the present paper is different in purpose from that of F. T. Barwell, as also recognized by the discussors. It is not only a matter of terminology: the main difference appears to be that Barwell was concerned with a summary of wear phenomena, irrespective of its relation to failure of any mechanism, whereas the present paper is a systematic overview of the failure modes in rolling contact mechanisms.

The author does believe, as suggested by the discussors, that under given operating conditions, more than one of the available failure modes can compete within a given bearing. By this is meant that more than one failure mode can act at the same time, and, depending on the conditions, one of them will be first to render the mechanism unserviceable. Of course, the fact that the failure modes compete does not mean that they cannot interact and reinforce each other, as do, for example, surface fatigue and spalling fatigue.

West and Moyer are adherents of Way's theory of crack propagation by the penetration of lubricant into cracks open to the surface. For a variety of reasons, this author is of the opinion that the Way theory is an unlikely explanation of the phenomena that take place in crack propagation. It so happens that the discussors have pointed to an interesting

argument *against* the Way theory when they mention that, according to that theory, surface fissures (furrows) should present exit paths for entrapped lubricant in a crack so that detrimental hydraulic pressure build-up is reduced, and therefore failures should not occur in the vicinity of furrows. Extensive evidence has been gathered lately by Martin (C3) that surface furrows are favored sites of spall initiation, and not unfavorable sites. Inasmuch as the discussors' conjecture holds that furrows tend to prevent operation of the pressure mechanism postulated by Way, the frequency of spalls associated with furrows can only mean that the Way mechanism does not operate in general.

The author is grateful for being reminded that initiation and propagation periods of rolling contact fatigue failures are so very different in their relative length when comparing through-hardened and case-hardened rolling bearing parts. It is quite true that in case-hardened material, spalls propagate very slowly, so much so that at times propagation seems to be halted for long periods. It is equally true that spall propagation is more rapid in both ball and roller bearings made of through-hardened steels. For this reason, there must be a difference in the metallurgy of the two types of material that account for the difference of propagation rates. It has often been suggested that this difference is the high residual compressive stresses present in the case of carburized parts.

The author agrees with West and Moyer that the ASA bearing rating standards apply to spalling fatigue failures only. It is also agreed that the rating method, whether applied to through-hardened or case-hardened elements, predicts the *initiation* of the spall rather than the ultimate loss of serviceability of the part. For some practical purposes, it is permissible to operate a rolling bearing in the presence of a spall as long as there is no catastrophic fracture or other gross unserviceability, whereas in other practical cases, this is impermissible due to the high vibrations set up or other undesirable phenomena. As the discussors state, it is up to the practicing engineer to select a failure criterion to suit his purpose, based on an understanding of the definition of life to failure comprised in the rating standards.

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